

CeNTREX short interaction region — EDM (Schiff-moment) signal simulation

Summary of the method, the simulation chain, the noise sources, and what they imply for the achievable CP-odd frequency sensitivity, data-taking time and magnetic shielding.

Code: `Small_IR_EDM_simulation/short_ir/` (Python package) and `short_ir_monte_carlo.ipynb`. Beam, switching and photon-cycling models are reused from `centrex_asymmetry` (`centrex_beam`); its authoritative production path is `run_full_16block_fluorescence.py` + `apply_saved_fluorescence.py`.

All numbers below were measured with this code unless marked as taken from a paper.

1. Idea in one paragraph

The short IR (~2 m of field, 5.2 m source-to-detector) uses the same E field (29.7 kV/cm), state preparation and detection as the main IR, but is shorter and less well shielded. We simulate a Ramsey measurement on one pair of hyperfine levels ($j \leftrightarrow e$, 119.52 kHz) using **Emma's COMSOL RF and E field maps integrated exactly**, instead of the idealised two-square-pulse Ramsey formula, and feed the result into `centrex_asymmetry`'s beam Monte Carlo, three-switch scheme and explicit LIF photon-cycling readout. A CP-odd frequency f_{CP} is injected, and extracted exactly as an experiment would. The scatter of the extracted value over many simulated runs is the sensitivity; any offset is a systematic.

2. Physics model

2.1 Two-level Ramsey through the real field

$$H(z) = \frac{1}{2} \begin{pmatrix} -\delta(z) & \Omega^*(z) \\ \Omega(z) & \delta(z) \end{pmatrix}, \quad \Omega(z) = 2\pi M(E(z)) [B_1(z) + B_2(z)e^{i\phi_2}]$$

$$\delta(z) = 2\pi [\nu_{\text{RF}} - \nu_{je}(E(z)) - \frac{\partial \nu}{\partial B_{\parallel}} B_{\parallel} - \Delta \nu_{\perp} - \nu_{\text{switch}} - \nu_{\text{mot}}(z, v)]$$

- Coil 2 is coil 1 mirrored about the domain centre; the two coils overlap (peaks at 0.803 and 1.197 m, 0.394 m apart), so this is not a clean two-pulse sequence.
- $M(E)$ and $\nu_{je}(E)$ are looked up from tables along z , so the fringe-field region (where ν_{je} moves by up to 1.3 kHz) is included.
- Exact 2×2 propagator per z step. The generator depends only on z and the time step is dz/v , so the whole velocity grid is propagated in one pass (1500 velocities × 4000 steps ≈ 0.1 s). $P(v)$ is tabulated once per switch state and interpolated.
- Operating point: the coil-2 phase is solved so that $P_e = 0.5$ (zero crossing, maximum slope) at 184 m/s — 1.6831 rad at Emma's amplitude, not exactly $\pi/2$ — then frozen.

2.2 Key parameters

quantity	value	source
RF amplitude <code>FIELD_SCALE</code>	0.365 (Emma); 0.3828 recommended (§5)	Emma / this work
E plateau	29.72 kV/cm	COMSOL
beam velocity	184 ± 16 m/s within a shot	centrex_asymmetry
shot-to-shot mean-velocity scatter	4 m/s	centrex_asymmetry
source start-time spread	1 ms	centrex_asymmetry
flight length	5.2 m (mean arrival 28.3 ms)	short IR
interaction time T	≈ 2.4 ms (2.0–2.9 ms across arrival bins)	measured
$B_{\parallel}$ (along E)	0.45 mG → −1.119 Hz	measured record
$B_{\perp}$ (static)	0.80 mG (⊥ v and E), 7.40 mG (along beam) → +0.0115 Hz	measured record
$\partial\nu/\partial B_{\parallel}$	−2486.5 Hz/G (= 2.49 mHz/μG, as CeNTREX)	centrex_tlf
κ (2nd-order transverse Zeeman)	207.5 Hz/G ²	from the two lines above

Axis note: the field record's position is quoted as $z = 44\text{--}88$ cm, so its z was taken as the beam axis and its y as the axis perpendicular to both v and E . This is inferred.

2.3 Magnetic effects included

effect	how it enters	parity under E reversal
static $B_{\parallel}$, first-order Zeeman	constant detuning, same in all switch states	even → rejected
static transverse field, 2nd order	constant detuning $\kappa \ B_{\perp}\ ^2$	even → rejected
slow one-directional $B_{\parallel}$ drift	$B_{\parallel}$ stepped between runs (1 run = 1 day); default off	even → rejected
motional field $B_{\text{mot}} = \mathbf{v} \times \mathbf{E}/c^2$	$\kappa[(b_{\perp} + B_{\text{mot}})^2 - b_{\perp}^2]$ per z step and velocity; sign follows the CP switch	odd → fake EDM

$B_{\text{mot}} = 60.8 \mu\text{G}$ at 184 m/s. The E-odd cross term $2\kappa b_{\perp} B_{\text{mot}}$ gives **−25.1 nHz per μG of $b_{\perp}$** (half the ±E difference; 50.2 nHz/μG for the difference, matching the CeNTREX proposal's ≈50 nHz/μG, arXiv:2010.01451 §3.3.2), proportional to v .

3. Simulation chain

step	what happens	code
1. field	load COMSOL CSVs (explicit header=), interpolate on a 4000-point z grid	<code>fields.py</code>
2. operating phase	solve $P_e = 0.5$ at 184 m/s, freeze	<code>ramsey.find_operating_phase</code>
3. switch tables	one $P(v)$ table per switch state (d, p, c) : $\phi_2 = \phi_{\text{op}} + p \phi_{\text{cal}}$, $\nu_{\text{switch}} = d \Delta f_{\text{sw}} + c f_{\text{CP}}$, motional sign = c	<code>ramsey.build_switch_tables</code>
4. schedule	N_{blocks} blocks; each block = the 8 states once, random order	<code>centrex_beam.switching</code>
5. shot sampling	shot mean velocity $\sim \mathcal{N}(184, 4)$; counts per (velocity bin, arrival bin, spin) drawn directly (§3.1)	<code>fast_shot.simulate_shot_binned</code>
6. LIF readout	each 10 μs arrival bin $\rightarrow$ two alternating 5 μs laser windows; explicit photon cycling; populations estimated from detected photons	<code>fluorescence.py</code> $\rightarrow$ <code>photon_cycling</code>
7. bin-first extraction	per block, per 1 ms arrival bin: switch parities; per block, through-origin fit $A_c = r A_d$ across bins, $f_{\text{CP}} = r \Delta f_{\text{sw}}$	<code>run.extract_bin_first(_fluorescence)</code>
8. combination	plain mean over blocks; error = empirical block SEM	<code>run._fit_and_combine</code>

3.1 Fast binned sampler

Given v , spin and arrival time are conditionally independent, so the joint distribution of counts factorises exactly:

$$n_j \sim \text{Mult}(N, q_j), \quad N_{1j} \sim \text{Bin}(n_j, \bar{p}_j), \quad \{c_{1jk}\}, \{c_{2jk}\} \sim \text{Mult}(N_{1j}, w_{jk}), \text{Mult}(N_{2j}, w_{jk})$$

(q_j : velocity-bin mass by erf; $\bar{p}_j$: density-weighted bin average of $p(v)$; w_{jk} : arrival-bin mass with the within-bin flight-time spread folded into the start-time width). Only the within-bin spin–arrival correlation, $O(\Delta v^2)$, is lost. Cost is independent of N : **46 ms vs 4.4 s per 2×10^7 -molecule shot** (per-molecule reference); $E[A]$ agrees to 5.8×10^{-6} . Details: `docs/binned_sampler_explained.md`.

3.2 Three-switch estimators

$$A_{dpc} \simeq A_0 + G_A [p \phi_{\text{cal}} + d 2\pi \Delta f_{\text{sw}} T + c \phi_{\text{CP}}], \quad f_{\text{CP}} = \Delta f_{\text{sw}} \frac{A_c}{A_d}$$

G_A and T cancel in the ratio. $\Delta f_{\text{sw}} = 0.5$ Hz, $\phi_{\text{cal}} = \pi/32$. ACME analogues: $\tilde{c} \leftrightarrow \tilde{N}\tilde{E}$, $\tilde{d} \leftrightarrow \tilde{B}$ (known precession), $\tilde{p} \leftrightarrow \tilde{\theta}$ (contrast). Unlike ACME, the switches are not nested by speed; all three flip on a ~ 1.7 -shot timescale.

Why bin-first. Molecules in one arrival bin share a velocity, hence T and pulse area, so the linearisation holds bin by bin. Pooling the whole run instead carries the run's single CP-odd velocity imbalance δv_c straight into the answer (a fake slope across arrival time, and a ~ 4 mHz offset with a 0.04 mHz quoted error in one test run).

4. Noise sources

4.1 Statistical terms

source	origin	scales with molecules/shot
spin projection	Bernoulli outcome of the Ramsey measurement	$1/\sqrt{N}$
LIF photon readout	geometric number of photons per molecule (pump-out $q \approx 0.48$, ~ 2.07 photons) $\times$ binomial detection ($\eta = 5.2\%$; $\sim 90\%$ of molecules give no detected photon). Per-molecule relative variance $\delta = q(1/\eta - 1) + (1 - q) \approx 9.3$; σ_A inflated by $\sqrt{1 + \delta} = 3.20$ (measured 3.12)	$1/\sqrt{N}$
shot-to-shot velocity scatter	every molecule of a shot shares a mean-velocity offset, changing pulse area and T ; leaks into the EDM channel through $\delta v_c = \frac{1}{8} \sum c_i \bar{v}_{\text{shot},i}$	does not scale

Variance budget, $f_{\text{CP}} = 0$, 4-block runs, 10^7 molecules/shot, Emma's 0.365 (paired, 40 seeds): LIF readout **63%**, velocity scatter **30%**, spin projection **7%**.

Per-block scatter of f_{CP} (8-shot block, LIF readout, SD known to $\sim 7\%$):

	0.365 (Emma)	0.3828 (local null)	0.3828, no shot velocity scatter
10^7 molecules/shot	33.1 mHz	24.2 mHz	—
10^{10} molecules/shot	17.0 mHz	3.49 mHz	0.69 mHz

At 10^{10} the photon and spin terms have dropped by $\sqrt{1000}$ and the **residual (second-order) velocity scatter is $\sim 96\%$ of the variance**.

4.2 Dependence on the injected f_{CP}

$\sigma^2(f_{\text{CP}}) = \sigma_0^2 + (\kappa_m f_{\text{CP}})^2$: an additive floor, flat below ~ 0.1 Hz ($f_{\text{CP}} = 0$ and 3×10^{-7} Hz agree to 0.1%), plus a multiplicative term from the noise on the ruler A_d , dominant above ~ 0.4 – 0.6 Hz. A physical signal ($\sim 10^{-7}$ Hz) sits on the floor, so **only $f_{\text{CP}} \lesssim 0.1$ Hz gives physical error bars**; the earlier $\pi/128 = 1.61$ Hz injection does not. The notebook uses 0.01–0.1 Hz; final sensitivity runs use 0.

4.3 Biases

- **Regression dilution** of the through-origin fit: slope attenuated by $\lambda = 1 - \sum w \text{var}_d / \sum w A_d^2$, $\propto 1/(\text{counts per bin})$. At 1 ms bins: -0.3% (projection) and **-3.2% with LIF** (measured -3.3% in a 20-run campaign). It is *multiplicative*, so it vanishes at $f_{\text{CP}} = 0$ and is irrelevant for a physical measurement; it matters only for calibration runs with a large injected signal.
- **Motional field**: $+20.1$ μHz at the measured $b_{\perp} = -0.80$ mG (j/e only). Additive, does not average down.
- **$B_{\parallel}$ drift** at the measured 0.0667 mG/day: walks the fringe offset A_0 by -2.4×10^{-3} /day exactly as predicted, but leaves f_{CP} and its error unchanged (common mode). At $10\times$ the rate the working point leaves the zero crossing (contrast $0.98 \rightarrow 0.89$, error $+50\%$): a sensitivity loss curable by daily re-tuning, not a fake EDM.

5. RF amplitude optimisation

At 0.365 the peak of $P_e(v)$ is at 175 m/s, not 184, so the asymmetry responds to the shot velocity at first order. Two nulls:

null	criterion	FIELD_SCALE	relevant for
local	$dA/dv = 0$ at 184 m/s	0.3828 (+4.9%)	bin-first (what we use)
shot-averaged	$d\langle A \rangle_{\text{shot}}/d\bar{v}_{\text{shot}} = 0$	0.3747 (+2.7%)	pooled estimator

Paired MC (same beam at every amplitude), coupling of the per-block f_{CP} to δv_c : +12.8 mHz/(m/s) ($r = 0.59$) at 0.365 $\rightarrow$ -0.4 mHz/(m/s) ($r = -0.03$) at 0.3828. Error bar $0.73\times$ at 10^7 ($\approx 54\%$ of the beam time), **$0.21\times$ at 10^{10} ($\approx 4\%$ of the beam time)**. Contrast is unchanged. The more molecules per shot, the more the amplitude matters.

6. Sensitivity expectations

6.1 Data-taking time

Blocks are independent (campaign Birge ratio ≈ 1), so

$$T = \frac{8}{R\eta_{\text{duty}}} \left(\frac{\sigma_{\text{block}}}{\sigma_{\text{target}}} \right)^2$$

with σ_{block} from §4.1. For 10^{10} molecules/shot, $R = 50$ Hz, **90% duty cycle** (one block = 0.18 s):

target $\sigma(f_{\text{CP}})$	0.3828, 4 m/s scatter	0.365, 4 m/s scatter	0.3828, no shot velocity scatter
10 mHz	0.2 s (1 block)	0.5 s	0.2 s
1 mHz	2.2 s	51 s	0.2 s
100 μ Hz	3.6 min	1.4 h	8 s
10 μ Hz	6.0 h	6 d	14 min
1 μ Hz	25 d	1.6 yr	23 h
300 nHz	278 d	18 yr	11 d

6.2 Magnetic shielding (motional field only)

Requirement: fake shift $\leq$ target, $b_{\perp,\text{max}} = \sigma_{\text{target}}/k$, with $k = 25.1$ nHz/ μ G (j/e only, this model) or 2.8 nHz/ μ G with CeNTREX's M modulation (e $\leftrightarrow$ j vs h $\leftrightarrow$ k; suppression by the proposal's ratio 5.6/50, independent of convention). Independent of molecule number.

target	$b_{\perp,\text{max}}$, j/e only	$b_{\perp,\text{max}}$, with M
10 mHz	400 mG	3.6 G
1 mHz	40 mG	360 mG
100 μ Hz	4.0 mG	36 mG
10 μ Hz	0.40 mG	3.6 mG
1 μ Hz	40 μ G	0.36 mG
300 nHz	12 μG	107 μG

With today's $b_{\perp} = 0.8$ mG at 10^{10} molecules/shot the measurement becomes systematics-limited after **~1.5 h** (j/e only, 20 μ Hz) or **~5 days** (with M, 2.2 μ Hz).

6.3 Reading the tables

- ≥ 100 μ Hz: current shielding suffices; minutes to hours of data.
- 10 μ Hz: ~6 h; needs M modulation, or shielding to 0.4 mG.
- 1 μ Hz: ~25 d; shielding to 0.36 mG even with M (today $\times 2$ too high).
- 300 nHz (main-IR scale): ~9 months, limited by **shot-to-shot velocity scatter** (removing it: 11 d), and shielding to ~0.1 mG with M (today $\times 7.5$ too high).

6.4 Comparison with the main IR

Normalised per shot and molecule, the short IR is **5.60 $\times$ worse** than the main-IR production run (block SEM 4.84×10^{-4} Hz over 128 shots at 7×10^7), against an interaction-time ratio of 5.61 $\times$: no penalty beyond the shorter T , i.e. **31 $\times$ the data** for the same precision. CeNTREX projects ≈ 90 nHz for the main IR in 300 h at 50 Hz (arXiv:2010.01451).

7. Validation performed

- Two independent codes (this package vs Emma's `ramsey4matrix.py`, after fixing its file names and a pandas header bug) agree to 3.7×10^{-15} .
- Binned vs per-molecule sampler: $E[A]$ to 5.8×10^{-6} , realisations within shot noise.
- LIF: calibrated totals within 0.07% of truth; residence 14 windows, inflation 3.2020 and zero carry-over, as in the main-IR path; paired readout inflation 3.12 vs 3.20.
- Variance budget closes to -7.6% (within the 11% SD uncertainty).
- Motional: coupling linear from 1 μ G to 100 mG; propagated/analytic 0.994; $\pm E$ difference 50.2 vs the proposal's ≈ 50 nHz/ μ G. Full-chain IPV (± 0.4 G): -16.3 ± 4.4 vs -25.1 nHz/ μ G — right sign, 2σ low, unresolved (re-run at 10^{10} molecules for a $7\times$ sharper check).
- Common-mode B drift: A_0 slope matches prediction to $< 1\%$.

8. Not modelled / open items

- **Other E-odd magnetic channels**, so the shielding numbers are necessary, not sufficient: leakage-current field (needs $\lesssim 1$ nA, independent of shielding), $B^{\text{nr}} \times \mathcal{E}^{\text{nr}}$ (needs an E-dependent Zeeman coefficient and an asymmetric E reversal), field gradients $\times$ trajectory (needs transverse motion).
- **M modulation** (second transition pair $h \leftrightarrow k$): only its suppression factor is used.
- **8-level manifold**: out of scope; the pulse pull ($+0.16$ Hz, $\propto B_{\text{RF}}^2$) is common to all switch states.
- **Background light, laser-power and detector-efficiency drifts**: absent; background adds to the photon noise and does not shrink with molecule number.
- **Molecule number per shot, duty cycle, rep rate** are assumptions; the times in §6.1 scale as $1/(N\eta)$ only while photon/spin noise dominate.
- **Shot-to-shot velocity scatter** (4 m/s) is taken from `centrex_asymmetry`; at high molecule number it is the dominant term, so its measured value matters most.
- Transverse-field drift, non-monotonic (e.g. diurnal) drift, drift within a run.
- Axis assignment of the measured field ($b_{\perp} = B_y$) is inferred.

9. Code map

file	contents
short_ir/config.py	all parameters (ShortIRConfig), drift (drifted), motional-field settings
short_ir/fields.py	COMSOL loading, mirroring, $M(\mathbf{E})$, $\nu_{je}(\mathbf{E})$, detuning
short_ir/ramsey.py	velocity-vectorised propagator (with motional term), RamseyTable, switch tables
short_ir/fast_shot.py	binned and per-molecule shot samplers
short_ir/fluorescence.py	LIF readout via centrex_beam.photon_cycling
short_ir/run.py	block runner (shot_mean_seed for paired beams), pooled and bin-first extraction

Notebook short_ir_monte_carlo.ipynb: setup and operating phase → $P(\mathbf{v})$ vs the analytic model → sampler validation → three-switch run → per-shot asymmetry and switching-block diagram → pooled extraction → bin-first extraction (with its own four-panel view) → RF amplitude optimisation → motional field (deterministic; IPV check behind RUN_MOTIONAL_IPV) → open items → multi-day campaign (LIF, local-null amplitude, optional drift diagnostics).